

**BUDDYAI JOB RECOMMENDATION SYSTEM
FOR CANDIDATE**

Capstone-II project report submitted

by the student of

Hybrid UG program in Computer Science & Data Analytics

Student Name-Uditya Seth

Roll No-2312res962

Group No-129

INDIAN INSTITUTE OF TECHNOLOGY PATNA

BIHTA - 801106, INDIA

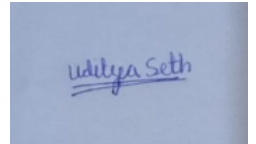
Date-29/05/2026

Declaration

I hereby declare that this submission is my own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Date:
29/05/2026

Student Name, Roll No, GroupNo, Signature.
Uditya Seth, 2312res962, 129 ,

A rectangular box containing a handwritten signature in blue ink. The signature reads "Uditya Seth" with a horizontal line underneath the name.

Summary of the Project

The BuddyAI Job Recommendation System is a system created to assist people who are searching for jobs by providing them with the best possible options based on their skills, qualifications, and experience. It combines a number of components into a single platform, providing a complete solution for resume analysis and recommendation.

One of the major features of this project is the AI Resume Analyzer, which is powered by the Google Gemini API. Unlike traditional Natural Language Processing (NLP) approaches, the system uses the Google Gemini API to intelligently analyze a resume and fetch valuable information from it. It also provides a number of detailed analyses, including strength, weakness, and improvement suggestions to help users improve their resume effectively.

To build a robust dataset of jobs, more than 1000 job listings were gathered from a number of online job portals by applying the technique of web scraping. Libraries such as BeautifulSoup and Selenium are used to fetch essential information from the websites. After collecting the data, Pandas was used for data analysis.

For data management and storage, MongoDB was used as a database. It provides efficient management of data related to jobs and users. It was also integrated into the system for the job recommendation system.

The AI Resume Analyzer was also created using the Streamlit library. It provides a powerful interface for users. All components of the system are integrated into a single platform, making it efficient and easy to use.

The BuddyAI system has demonstrated how modern AI tools and data technologies can be combined to simplify the job search process, making it smarter, faster, and more user-centric.

Mention the work done by each member:

1 Uditya Seth – Data Scrapping of Job Data And Formatting Data in Structured way According to Data Base And Build AI Resume Analyzer Which Is Hosted Through Streamlit And Add on Features Like Mock Test and Chat Bot

2 Jha Shivam Shambhunath – He Worked in A Backend Of BuddyAI And Also Make Job Matcher And Also Help in Adding Features Like Resume Uploader , AI Mock Interview , Mock Analytics

3 Alquama Shaibli – He Worked On UI Part He Make One Web Page For My Resume Analyzer According To Website And He Done with Frontend Part of Job Portal

4 Aqdas Shahid – He Didn't Contribute In Our Project And Never Respond In Group

5 Subhash Sharan - He Didn't Contribute In Our Project And Never Respond In Group

Contents

- 1 Introduction
- 2 Problem Statement
- 3 Objectives
- 4 Literature Survey
- 5 Proposed System
- 6 System Architecture
- 7 Module Description
- 8 Technology Stack
- 9 Screenshots
- 10 Screenshots
- 11 Conclusion & Future Scope
- 12 References

Chapter – 1

Introduction

In the modern digital era, the job market has become highly competitive and dynamic. Candidates often struggle to identify job opportunities that align with their skills, while recruiters face challenges in efficiently filtering and selecting the most suitable candidates. Traditional job search methods are time-consuming and often inefficient, leading to mismatches between job requirements and candidate profiles.

To address these challenges, our project “AI Job Buddy” aims to develop an intelligent job recommendation system that leverages Artificial Intelligence to bridge the gap between job seekers and recruiters. The system integrates job data collection, resume analysis, and job matching into a single platform.

A key component of this system is the AI Resume Analyzer, which plays a crucial role in extracting meaningful information from resumes. It analyzes candidate data such as skills, experience, education, and keywords, and compares it with job descriptions to determine compatibility. This helps in identifying the most suitable job opportunities for each candidate. In addition to resume analysis and job matching, the system now incorporates mock test modules and an AI-powered chatbot assistant, further enhancing candidate preparation and guidance.

Chapter – 2

Problem Statement

Recruiters face challenges such as:

- Manual screening is time-consuming.
- Keyword-based searches fail to capture semantic meaning.
- Lack of personalized recommendations.
- Difficulty in handling large datasets.

Thus, there is a need for an automated system that can intelligently analyze resumes, match them with job postings, and provide actionable insights.

Chapter 3

Objectives

- Scrape job postings from multiple portals to build a dataset.
- Develop an AI Resume Analyzer using Gemini.
- Deploy the analyzer via Streamlit for interactive use.
- Build backend APIs and database integration.
- Implement a job matcher algorithm.
- Design a web interface for usability.
- Develop mock test modules for both technical and non-technical roles (BBA, MBA, etc).
- Integrate a chatbot assistant using Gemini to provide instant support for resume queries, keyword explanations, and job description clarification
- Develop a Resume Uploader for seamless candidate resume submission.
- Implement AI Mock Interview modules to simulate real interview scenarios.
- Build Mock Analytics to provide candidates with detailed performance feedback

Chapter 4

Literature Survey

Existing systems like LinkedIn, Indeed, and Naukri provide job listings and resume uploads but lack advanced AI-driven analysis. Recruiters often rely on keyword searches, which do not capture context.

Research shows that AI models can improve recruitment efficiency by:

- Extracting skills and keywords.
- Performing semantic similarity analysis.
- Providing personalized recommendations.
- Assessment modules
- Conversational AI assistants

Limitations of current systems:

- Manual screening is slow.
- Limited semantic analysis.
- Lack of recruiter dashboards.
- No integrated mock testing or AI-driven assistance for resume and JD clarification.

Chapter 5

Proposed System

The proposed system integrates multiple modules:

- **Data Scraping:** Collects job postings.
- **Resume Analyzer:** Uses Gemini AI to analyze resumes.
- **Backend:** Handles data flow and APIs.
- **Job Matcher:** Matches resumes with job postings.
- **Mock Test Module :** 11 role-specific tests for technical and non-technical domains (e.g., CS, IT, BBA, MBA).
- **Chatbot Assistant :** Gemini API-based conversational support for keyword clarification, JD explanation, and resume guidance.
- **Frontend Integration:** Streamlit UI hosting Resume analyzer, mock tests, and chatbot
- **Resume Uploader :** Allows candidates to upload resumes directly into the system for analysis.
- **AI Mock Interview :** Simulates interview sessions using Gemini AI, providing real-time feedback.
- **Mock Analytics :** Tracks candidate performance in mock tests and interviews, offering detailed insights and improvement suggestions.

This modular approach ensures scalability, personalization, and candidate readiness, positioning BuddyAI as a more comprehensive solution compared to existing platforms.

Chapter 6

System Architecture

The architecture consists of:

- 1. Data Scraping Module → Collects job data.**
- 2. Database → Stores job postings.**
- 3. Backend APIs → Connects modules.**
- 4. Resume Analyzer → Processes resumes using Gemini.**
- 5. Job Matcher → Compares resumes with jobs.**
- 6. Frontend Web Page → Displays results.**
- 7. Mock Test Module → Provides role-specific tests (technical + non-technical)**
- 8. Chatbot Assistant → Built with Gemini API, handles queries about resumes, JD explanations, and keyword guidance.**

Resume upload → Analyzer → Matcher → Mock Tests → AI Mock Interview → Mock Analytics → Chatbot → Results

Chapter 7

Module Description

- **Data Scraping (My Contribution):** Scraped 1000 job postings using Python (BeautifulSoup, Pandas and Data Bars). Cleaned and standardized data.
- **Resume Analyzer (My Contribution):** Built with Gemini AI. Extracts keywords, matches skills, and highlights strengths/weaknesses. Deployed via Streamlit.
- **Mock Test Feature(My Contribution):**Designed and implemented **11 mock tests** covering technical roles (CS, IT) and non-technical roles (BBA, MBA).
- **Chat Bot Fearture(My Contribution):**Built JobBuddy AI Chatbot using Gemini API integration to answer candidate queries beyond static analysis.
- **Backend Development (Teammate):** APIs for resume uploads and job data. Database integration for scalability.
- **Job Matcher (Teammate):** Algorithm ranks jobs based on skill similarity. Provides recommendations.
- **Web Page Development (Teammate):** Designed frontend for Job Portal.
- **Resume Uploader (Teammate):** Developed backend functionality for candidates to upload resumes securely.
- **AI Mock Interview (Teammate):** Designed interview simulation with Gemini AI, providing adaptive questions and feedback.
- **Mock Analytics (Teammate):** Implemented analytics dashboards to evaluate candidate performance in tests and interviews.

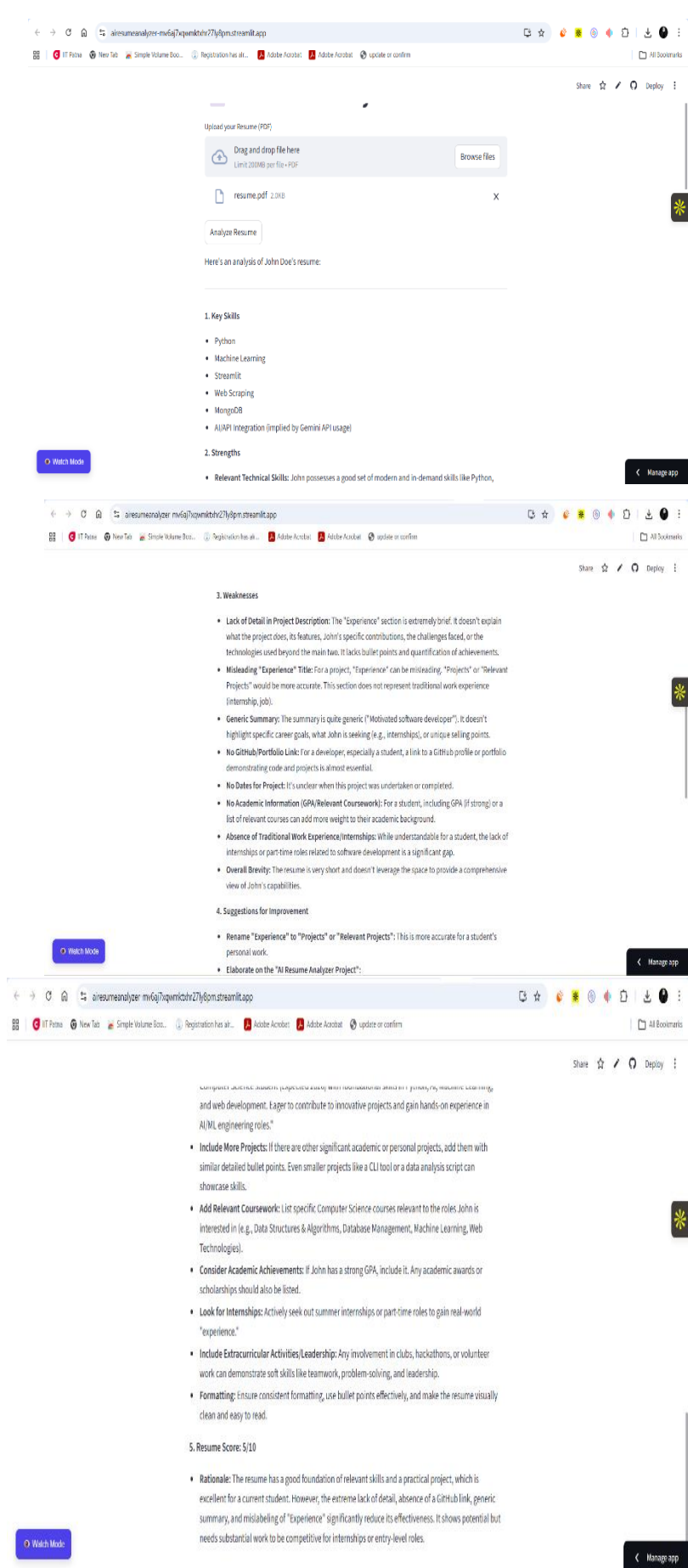
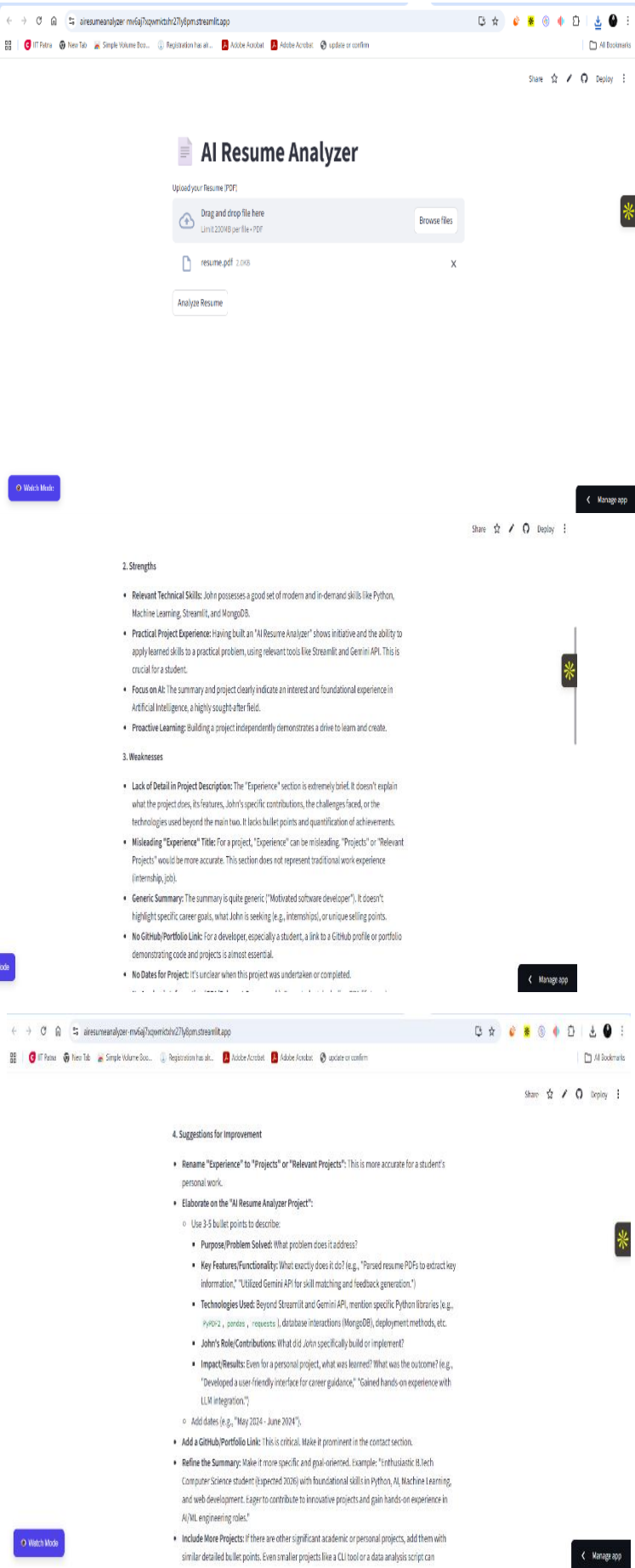
Chapter 8

Technology Stack

- Languages: Python.
- Libraries: BeautifulSoup, Pandas, NumPy, Streamlit
- AI Tool: Data Bar
- AI Model: Gemini.
- Database: Mongo DB
- Deployment: Streamlit for Resume Analyzer.
- Mock Test Framework: Python + Html + CSS +NodeJs
- Chatbot Integration: Java Script + Html + CSS
- Frontend Enhancement : Html + CSS + AI
- Resume Uploader: Python + Flask APIs
- AI Mock Interview: Gemini AI
- Mock Analytics: **Frontend:** React
Backend: Node.js + Express.js

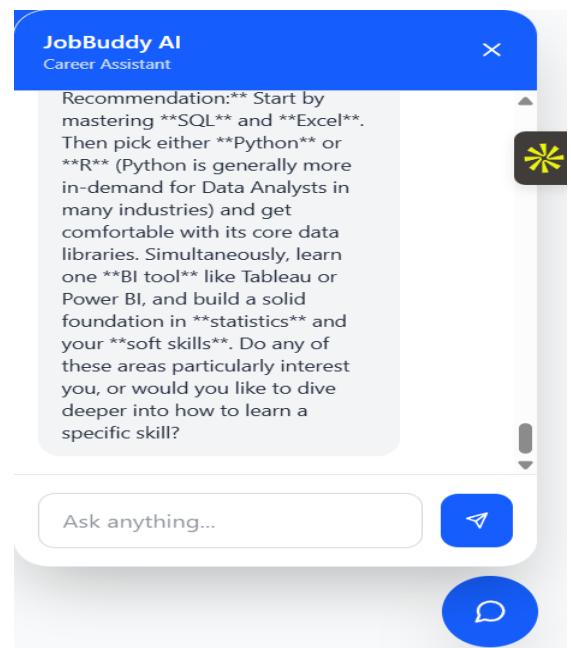
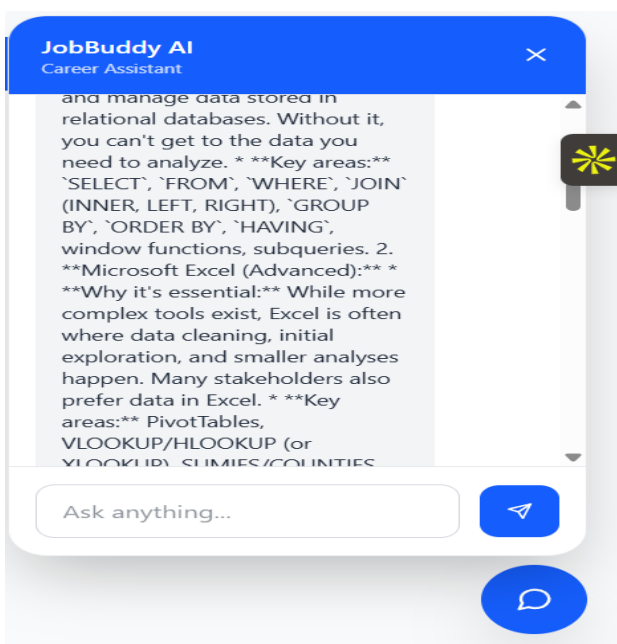
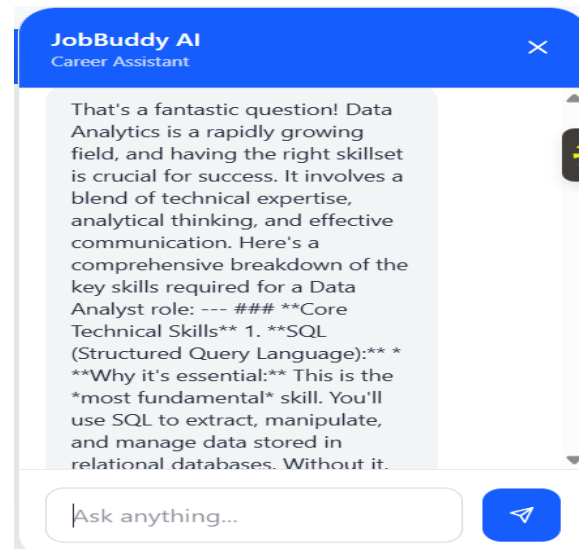
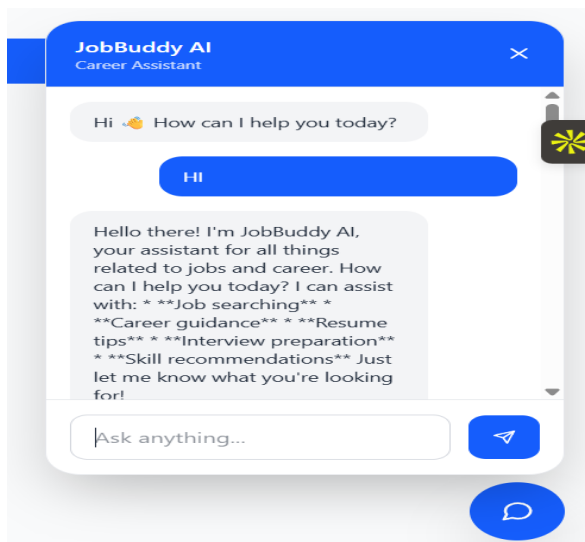
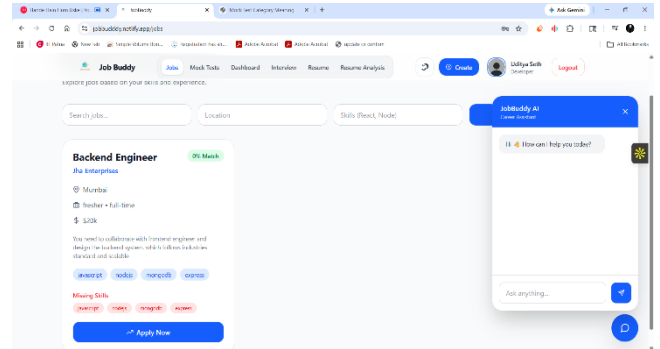
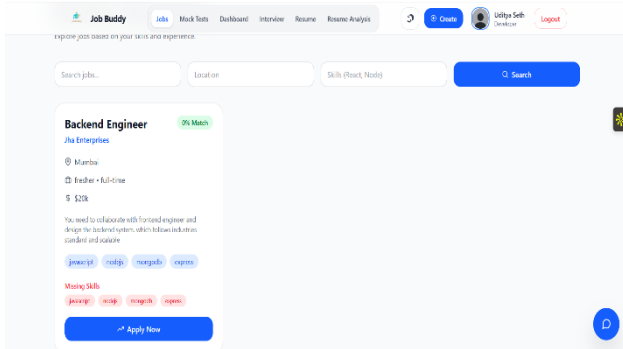
Chapter 9

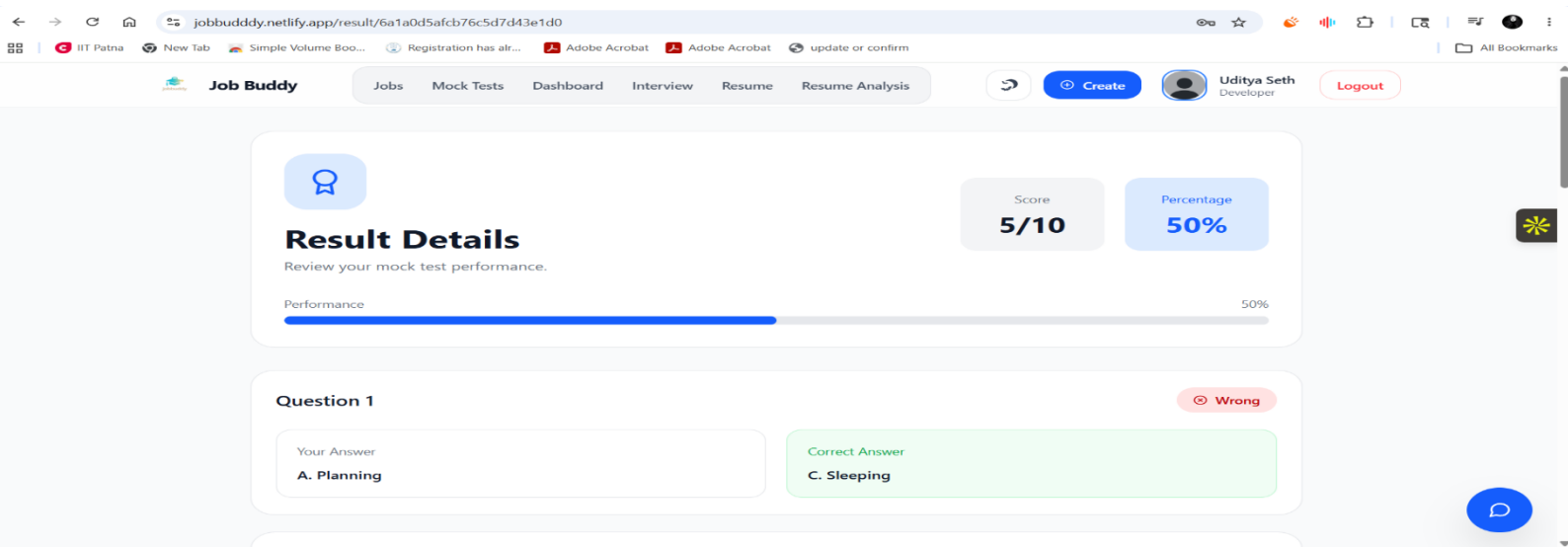
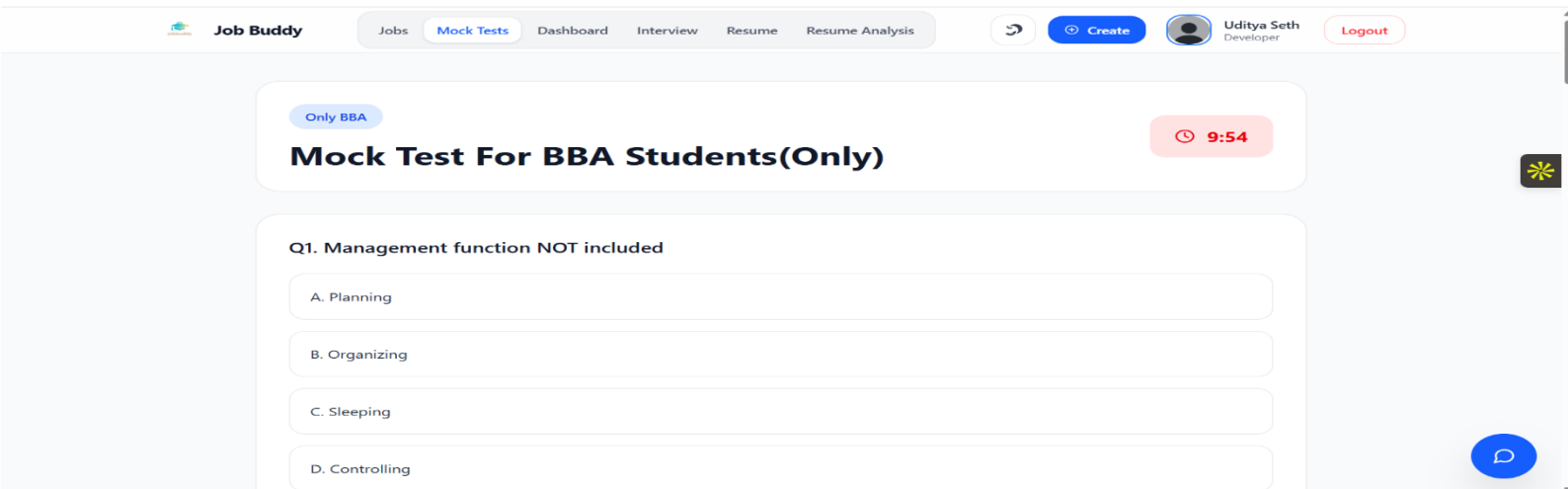
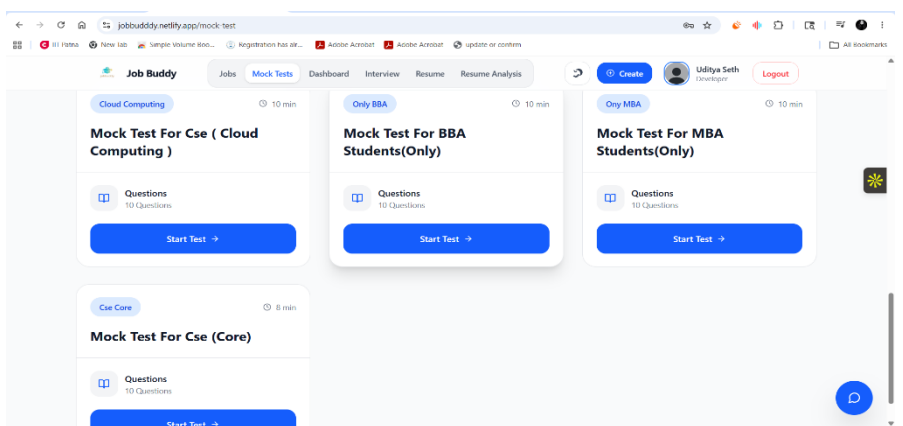
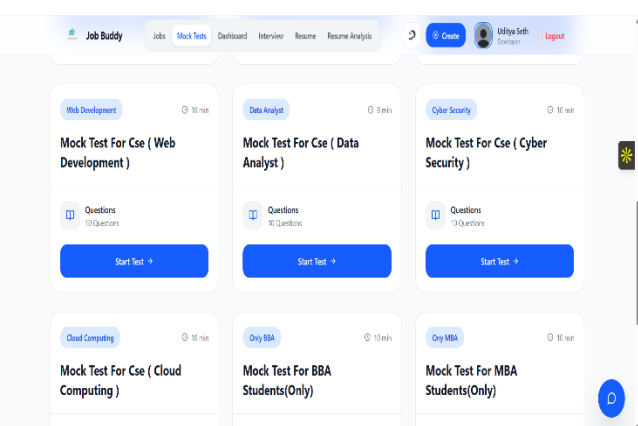
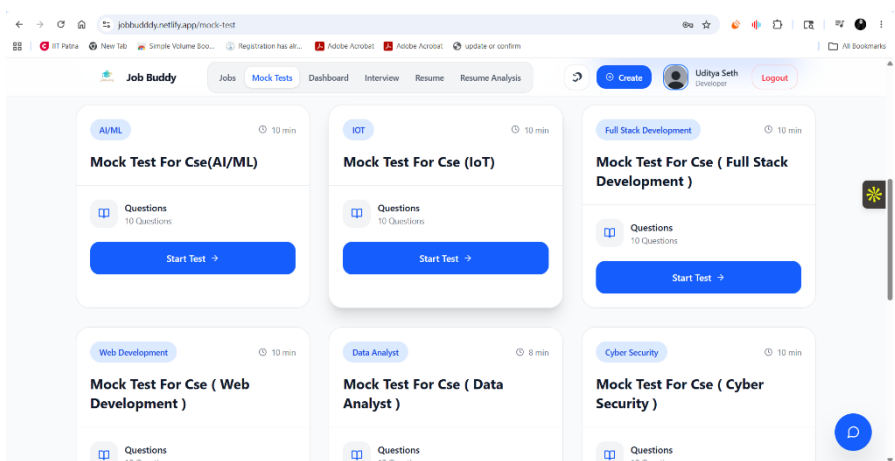
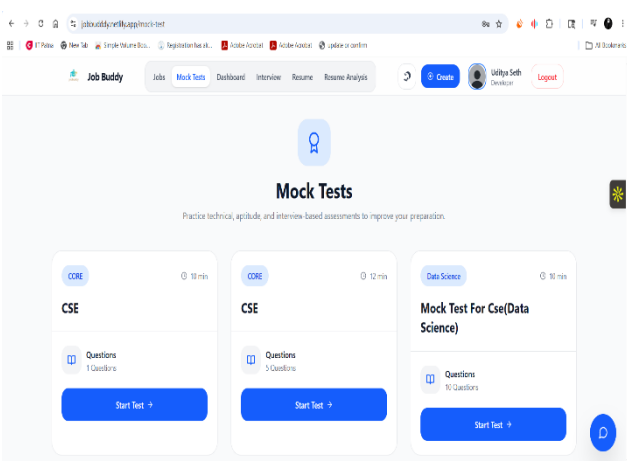
Screenshots



Chapter 10

Screenshots





Chapter 11

Conclusion & future work

JobBuddy addresses key challenges faced by job seekers by integrating job search, skill analysis, and resume improvement into one platform. This helps users improve their job readiness and provides a strong foundation for future enhancements.

Buddy AI now not only recommends jobs but also prepares candidates through mock tests and provides real-time assistance via an AI chatbot.”

Future Scope:

- Integration of AI-based mock interview system
- Real-time feedback on interview performance
- Personalized learning paths
- Adaptive learning paths based on mock test performance.
- Chatbot integration with recruiter dashboards for candidate Q&A.
- Expansion to industry-specific mock tests (finance, healthcare, etc.).
- Expansion of Mock Analytics into recruiter dashboards.
- Integration of AI Mock Interview with real-time recruiter feedback.

Chapter 12

References

- LinkedIn, Indeed, Naukri job portals.
- Python libraries: Beautiful Soup, Pandas Numpy.
- Streamlit Documentation.
- Chrome Extension - Data Bar
- Gemini AI model Documentations.
- Books – Java , Java Script
- Nodjs Documentation